we have $[\Phi(d_p + b_p^{-1}x)]^{p-1} \to e^{-e^{-x}}$. Hence it remains to show that

$$\frac{p}{b_p}\phi(d_p + b_p^{-1}x) \to e^{-x}.$$

Taking the logarithm of the left side yields

$$\log p - \log b_p - \log(\sqrt{2\pi}) - (d_p + b_p^{-1}x)^2/2. \tag{8}$$

Expanding $(d_p + b_p^{-1}x)^2$ gives that

$$d_p^2 + 2d_p b_p^{-1}x + b_p^{-2}x^2 = b_p^2 - \log\log p - \log(4\pi) + 2x + o(1),\ p \to \infty,$$

by which we have (8) $= -x + o(1)$. This shows that $g_p(x) \to g(x)$ for all $x \in \mathbb{R}$. Moreover, this convergence takes place locally uniformly in x, that is, for every $K > 0$, $g_p(x) \to g(x)$ uniformly in $x \in [-K, K]$.

On the other hand, the density of $\max_{1\le j\le p} X_j$ is given by $f_p(x) = p\phi(x)[\Phi(x)]^{p-1}$. By this form, for every $K > 0$, there exist a constant $c > 0$ and a positive integer p_0 depending only on K such that for $p \ge p_0$,

$$\inf_{x\in[d_p - Kb_p^{-1}, d_p + Kb_p^{-1}]} b_p^{-1} f_p(x) = \inf_{x\in[-K,K]} g_p(x) \ge \inf_{x\in[-K,K]} g(x) + o(1) \ge c,$$

which shows that for $p \ge p_0$,

$$f_p(x) \ge cb_p,\ \forall x \in [d_p - Kb_p^{-1}, d_p + Kb_p^{-1}].$$

Therefore, we conclude that for $p \ge p_0$,

$$\mathrm{P}(|\max_{1\le j\le p} X_j - d_p| \le \epsilon) = \int_{d_p-\epsilon}^{d_p+\epsilon} f_p(x)dx \ge 2c\epsilon b_p,\ \forall \epsilon \in [0, Kb_p^{-1}].$$

By the Gaussian maximal inequality and Lemma 2.3.15 of [11], we have

$$\sqrt{\log p}/12 \le \mathbb{E}[\max_{1\le j\le p} X_j] \le \sqrt{2\log p}.$$

Hence, by the previous result, for every $K' > 0$, there exist a constant $c' > 0$ and and a positive integer p_0' depending only on K' such that for $p \ge p_0'$, $a_p \ge 1$ and

$$\mathcal{L}(\max_{1\le j\le p} X_j, \epsilon) \ge \mathrm{P}(|\max_{1\le j\le p} X_j - d_p| \le \epsilon) \ge c'\epsilon a_p,\ \forall \epsilon \in [0, K'a_p^{-1}].$$

□

4. Proofs for Section 2

4.1. **Proof of Theorem 1.** Here for a smooth function $f : \mathbb{R}^p \to \mathbb{R}$, we write $\partial_j f(z) = \partial f(z)/\partial z_j$ for $z = (z_1, \ldots, z_p)^T$. We shall use the following version of Stein's identity.

Lemma 2 (Stein's identity)**.** *Let $W = (W_1, \ldots, W_p)^T$ be a centered Gaussian random vector in $\mathbb{R}^p$. Let $f : \mathbb{R}^p \to \mathbb{R}$ be a C^1-function such that $\mathrm{E}[|\partial_j f(W)|] < \infty$ for all $1 \leq j \leq p$. Then for every $1 \leq j \leq p$,*

$$\mathrm{E}[W_j f(W)] = \sum_{k=1}^p \mathrm{E}[W_j W_k]\mathrm{E}[\partial_k f(W)].$$

Proof of Lemma 2. See Section A.6 of [27]; also [8] and [25]. □

We will use the following properties of the smooth max function.

Lemma 3. *For every $1 \leq j, k \leq p$,*

$$\partial_j F_\beta(z) = \pi_j(z), \quad \partial_j \partial_k F_\beta(z) = \beta w_{jk}(z),$$

where

$$\pi_j(z) := e^{\beta z_j} / \textstyle\sum_{m=1}^p e^{\beta z_m}, \ w_{jk}(z) := 1(j = k)\pi_j(z) - \pi_j(z)\pi_k(z).$$

Moreover,

$$\pi_j(z) \geq 0, \ \textstyle\sum_{j=1}^p \pi_j(z) = 1, \ \sum_{j,k=1}^p |w_{jk}(z)| \leq 2.$$

Proof of Lemma 3. The first property was noted in [7]. The other properties follow from a direct calculation. □

Lemma 4. *Let $m := g \circ F_\beta$ with $g \in C^2(\mathbb{R})$. Then for every $1 \leq j, k \leq p$,*

$$\partial_j \partial_k m(z) = (g'' \circ F_\beta)(z)\pi_j(z)\pi_k(z) + \beta(g' \circ F_\beta)(z)w_{jk}(z),$$

where π_j and w_{jk} are defined in Lemma 3.

Proof of lemma 4. The proof follows from a direct calculation. □

Proof of Theorem 1. Without loss of generality, we may assume that X and Y are independent, so that $\mathrm{E}[X_j Y_k] = 0$ for all $1 \leq j, k \leq p$. Consider the following Slepian interpolation between X and Y:

$$Z(t) := \sqrt{t}X + \sqrt{1-t}Y, \ t \in [0, 1].$$

Let $m := g \circ F_\beta$ and $\Psi(t) := \mathrm{E}[m(Z(t))]$. Then

$$|\mathrm{E}[m(X)] - \mathrm{E}[m(Y)]| = |\Psi(1) - \Psi(0)| = \left| \int_0^1 \Psi'(t) dt \right|.$$

Here we have

$$\begin{aligned}
\Psi'(t) &= \frac{1}{2} \sum_{j=1}^p \mathrm{E}[\partial_j m(Z(t))(t^{-1/2} X_j - (1-t)^{-1/2} Y_j)] \\
&= \frac{1}{2} \sum_{j=1}^p \sum_{k=1}^p (\sigma_{jk}^X - \sigma_{jk}^Y) \mathrm{E}[\partial_j \partial_k m(Z(t))],
\end{aligned}$$

where the second equality follows from applying Lemma 2 to $W = (t^{-1/2}X_j - (1-t)^{-1/2}Y_j, Z(t)^T)^T$ and $f(W) = \partial_j m(Z(t))$. Hence

$$\begin{aligned}\left|\int_0^1 \Psi'(t)dt\right| &\le \frac{1}{2}\sum_{j,k=1}^p |\sigma_{jk}^X - \sigma_{jk}^Y| \left|\int_0^1 \mathrm{E}[\partial_j\partial_k m(Z(t))]dt\right| \\ &\le \frac{1}{2}\max_{1\le j,k\le p}|\sigma_{jk}^X - \sigma_{jk}^Y| \int_0^1 \sum_{j,k=1}^p |\mathrm{E}[\partial_j\partial_k m(Z(t))]|\,dt \\ &= \frac{\Delta}{2}\int_0^1 \sum_{j,k=1}^p |\mathrm{E}[\partial_j\partial_k m(Z(t))]|\,dt.\end{aligned}$$

By Lemmas 3 and 4,

$$\sum_{j,k=1}^p |\partial_j\partial_k m(Z(t))| \le |(g''\circ F_\beta)(Z(t))| + 2\beta|(g'\circ F_\beta)(Z(t))|.$$

Therefore, we have

$$\begin{aligned}&|\mathrm{E}[g(F_\beta(X)) - g(F_\beta(Y))]| \\ &\le \Delta \times \left\{\frac{1}{2}\int_0^1 \mathrm{E}[|(g''\circ F_\beta)(Z(t))|]dt + \beta\int_0^1 \mathrm{E}[|(g'\circ F_\beta)(Z(t))|]dt\right\} \\ &\le \Delta(\|g''\|_\infty/2 + \beta\|g'\|_\infty),\end{aligned}$$

which leads to the first assertion. The second assertion follows from the inequality (1). This completes the proof. □

4.2. **Proof of Theorem 2.** The final assertion follows from the inequality $a_p \le \sqrt{2\log p}$ (see, for example, [27], Proposition 1.1.3). Hence we prove (2). We first note that we may assume that $0 < \Delta < 1$ since otherwise the proof is trivial (take $C \ge 2$ in (2)). In what follows, let $C > 0$ be a generic constant that depends only on $\min_{1\le j\le p}\sigma_{jj}^Y$ and $\max_{1\le j\le p}\sigma_{jj}^Y$, and its value may change from place to place. For $\beta > 0$, define $e_{p,\beta} := \beta^{-1}\log p$. Consider and fix a C^2-function $g_0 : \mathbb{R} \to [0,1]$ such that $g_0(t) = 1$ for $t \le 0$ and $g_0(t) = 0$ for $t \ge 1$. For example, we may take

$$g_0(t) = \begin{cases} 0, & t \ge 1, \\ 30\int_t^1 s^2(1-s)^2 ds, & 0 < t < 1, \\ 1, & t \le 0. \end{cases}$$

For given $x \in \mathbb{R}, \beta > 0$ and $\delta > 0$, define $g_{x,\beta,\delta}(t) = g_0(\delta^{-1}(t - x - e_{p,\beta}))$. For this function $g_{x,\beta,\delta}$, $\|g'_{x,\beta,\delta}\|_\infty = \delta^{-1}\|g_0'\|_\infty$ and $\|g''_{x,\beta,\delta}\|_\infty = \delta^{-2}\|g_0''\|_\infty$. Moreover,

$$1(t \le x + e_{p,\beta}) \le g_{x,\beta,\delta}(t) \le 1(t \le x + e_{p,\beta} + \delta),\ \forall t \in \mathbb{R}. \tag{9}$$

For arbitrary $x \in \mathbb{R}, \beta > 0$ and $\delta > 0$, observe that

$$\begin{aligned}
\mathrm{P}(\max_{1\le j\le p} X_j \le x) &\le \mathrm{P}(F_\beta(X) \le x + e_{p,\beta}) \le \mathrm{E}[g_{x,\beta,\delta}(F_\beta(X))] \\
&\le \mathrm{E}[g_{x,\beta,\delta}(F_\beta(Y))] + C(\delta^{-2} + \beta\delta^{-1})\Delta \\
&\le \mathrm{P}(F_\beta(Y) \le x + e_{p,\beta} + \delta) + C(\delta^{-2} + \beta\delta^{-1})\Delta \\
&\le \mathrm{P}(\max_{1\le j\le p} Y_j \le x + e_{p,\beta} + \delta) + C(\delta^{-2} + \beta\delta^{-1})\Delta, \quad (10)
\end{aligned}$$

where the first inequality follows from the inequality (1), the second from the inequality (9), the third from Theorem 1, the fourth from the inequality (9), and the last from the inequality (1). We wish to compare $\mathrm{P}(\max_{1\le j\le p} Y_j \le x+e_{p,\beta}+\delta)$ with $\mathrm{P}(\max_{1\le j\le p} Y_j \le x)$, and this is where the anti-concentration inequality plays its role. By Theorem 3, we have

$$\begin{aligned}
&\mathrm{P}(\max_{1\le j\le p} Y_j \le x + e_{p,\beta} + \delta) - \mathrm{P}(\max_{1\le j\le p} Y_j \le x) \\
&= \mathrm{P}(x < \max_{1\le j\le p} Y_j \le x + e_{p,\beta} + \delta) \le \mathcal{L}(\max_{1\le j\le p} Y_j, e_{p,\beta} + \delta) \quad (11) \\
&\le C(e_{p,\beta} + \delta)\sqrt{1 \vee a_p^2 \vee \log\{1/(e_{p,\beta} + \delta)\}} \le C(e_{p,\beta} + \delta)\sqrt{1 \vee a_p^2 \vee \log(1/\delta)}.
\end{aligned}$$

Therefore,

$$\begin{aligned}
&\mathrm{P}(\max_{1\le j\le p} X_j \le x) - \mathrm{P}(\max_{1\le j\le p} Y_j \le x) \\
&\qquad \le C\left\{(\delta^{-2} + \beta\delta^{-1})\Delta + (e_{p,\beta} + \delta)\sqrt{1 \vee a_p^2 \vee \log(1/\delta)}\right\}. \quad (12)
\end{aligned}$$

Take $a = a_p \vee \log^{1/2}(1/\Delta)$, and choose β and δ in such a way that

$$\beta = \delta^{-1}\log p \text{ and } \delta = \Delta^{1/3}(1 \vee a)^{-1/3}(2\log p)^{1/3}.$$

Recall that $p \ge 2$ and $0 < \Delta < 1$. Observe that $(\delta^{-2} + \beta\delta^{-1})\Delta \le C\Delta^{1/3}(1 \vee a)^{2/3}\log^{1/3} p, (e_{p,\beta} + \delta)(1 \vee a_p) \le C\Delta^{1/3}(1 \vee a)^{2/3}\log^{1/3} p$, and since $\delta \ge \Delta^{1/3}(1\vee a)^{-1/3}$, we have $\log(1/\delta) \le (1/3)\log((1\vee a)/\Delta)$. Hence the right side on (12) is bounded by $C\Delta^{1/3}\{(1\vee a)^{2/3}\log^{1/3} p + (1\vee a)^{-1/3}(\log^{1/3} p)\log^{1/2}((1\vee a)/\Delta)\}$. In addition, $(1 \vee a)^{-1}\log^{1/2}(1 \vee a)$ is bounded by a universal constant, so that

$$\begin{aligned}
&\text{right side of (12)} \\
&\qquad \le C\Delta^{1/3}\{(1 \vee a)^{2/3}\log^{1/3} p + (1 \vee a)^{-1/3}(\log^{1/3} p)\log^{1/2}(1/\Delta)\}.
\end{aligned}$$

The second term inside the bracket is bounded by $(\log^{1/3} p)\log^{1/3}(1/\Delta)$ as $(1\vee a)^{-1/3} \le (\log(1/\Delta))^{-1/6}$, and first term is bound by $(1\vee a_p)^{2/3}\log^{1/3} p + (\log^{1/3} p)\log^{1/3}(1/\Delta)$. Adjusting the constant C, the right side on the above displayed equation is bounded by $C\Delta^{1/3}\{1 \vee a_p^2 \vee \log(1/\Delta)\}^{1/3}\log^{1/3} p$.

For the opposite direction, observe that

$$\begin{aligned}\mathrm{P}(\max_{1\le j\le p} X_j \le x) &\ge \mathrm{P}(F_\beta(X) \le x) \ge \mathrm{E}[g_{x-e_{p,\beta}-\delta,\beta,\delta}(F_\beta(X))] \\ &\ge \mathrm{E}[g_{x-e_{p,\beta}-\delta,\beta,\delta}(F_\beta(Y))] - C(\delta^{-2}+\beta\delta^{-1})\Delta \\ &\ge \mathrm{P}(F_\beta(Y) \le x-\delta) - C(\delta^{-2}+\beta\delta^{-1})\Delta \\ &\ge \mathrm{P}(\max_{1\le j\le p} Y_j \le x - e_{p,\beta} - \delta) - C(\delta^{-2}+\beta\delta^{-1})\Delta.\end{aligned}$$

The rest of the proof is similar and hence omitted. □

5. Proof of Theorem 3

The proof of Theorem 3 uses some properties of Gaussian measures. We begin with preparing technical tools. The following two facts were essentially noted in [31, 32] (note: [31] and [32] did not contain a proof of Lemma 5, which we find non-trivial). For the sake of completeness, we give their proofs after the proof of Theorem 3.

Lemma 5. *Let $(W_1,\dots,W_p)^T$ be a (not necessarily centered) Gaussian random vector in $\mathbb{R}^p$ with $\mathrm{Var}(W_j)=1$ for all $1\le j\le p$. Suppose that $\mathrm{Corr}(W_j,W_k)<1$ whenever $j\ne k$. Then the distribution of $\max_{1\le j\le p} W_j$ is absolutely continuous with respect to the Lebesgue measure and a version of the density is given by*

$$f(x) = \phi(x)\sum_{j=1}^p e^{\mathrm{E}[W_j]x-(\mathrm{E}[W_j])^2/2}\cdot \mathrm{P}\left(W_k\le x, \forall k\ne j \mid W_j = x\right). \tag{13}$$

Lemma 6. *Let $(W_0,W_1,\dots,W_p)^T$ be a (not necessarily centered) Gaussian random vector with $\mathrm{Var}(W_j)=1$ for all $0\le j\le p$. Suppose that $\mathrm{E}[W_0]\ge 0$. Then the map*

$$x \mapsto e^{\mathrm{E}[W_0]x-(\mathrm{E}[W_0])^2/2}\cdot \mathrm{P}(W_j\le x, 1\le \forall j\le p \mid W_0 = x) \tag{14}$$

is non-decreasing on $\mathbb{R}$.

Let us also recall (a version of) the Gaussian concentration (more precisely, deviation) inequality. See, for example, [15], Theorem 7.1, for its proof.

Lemma 7. *Let $(X_1,\dots,X_p)^T$ be a centered Gaussian random vector in $\mathbb{R}^p$ with $\max_{1\le j\le p}\mathrm{E}[X_j^2]\le\sigma^2$ for some $\sigma^2>0$. Then for every $r>0$,*

$$\mathrm{P}(\max_{1\le j\le p} X_j \ge \mathrm{E}[\max_{1\le j\le p} X_j] + r) \le e^{-r^2/(2\sigma^2)}.$$

We are now in position to prove Theorem 3.

Proof of Theorem 3. The proof consists of three steps.

Step 1. This step reduces the analysis to the unit variance case. Pick any $x \geq 0$. Let $W_j := (X_j - x)/\sigma_j + x/\underline{\sigma}$. Then $\mathrm{E}[W_j] \geq 0$ and $\mathrm{Var}(W_j) = 1$. Define $Z := \max_{1 \leq j \leq p} W_j$. Then we have

$$\begin{aligned}\mathrm{P}(|\max_{1\leq j\leq p} X_j - x| \leq \epsilon) &\leq \mathrm{P}\left(\left|\max_{1\leq j\leq p} \frac{X_j - x}{\sigma_j}\right| \leq \frac{\epsilon}{\underline{\sigma}}\right) \\ &\leq \sup_{y\in\mathbb{R}} \mathrm{P}\left(\left|\max_{1\leq j\leq p} \frac{X_j - x}{\sigma_j} + \frac{x}{\underline{\sigma}} - y\right| \leq \frac{\epsilon}{\underline{\sigma}}\right) = \sup_{y\in\mathbb{R}} \mathrm{P}\left(|Z - y| \leq \frac{\epsilon}{\underline{\sigma}}\right).\end{aligned}$$

Step 2. This step bounds the density of Z. Without loss of generality, we may assume that $\mathrm{Corr}(W_j, W_k) < 1$ whenever $j \neq k$. Since the marginal distribution of W_j is $N(\mu_j, 1)$ where $\mu_j := \mathrm{E}[W_j] = (x/\underline{\sigma} - x/\sigma_j) \geq 0$, by Lemma 5, Z has density of the form

$$f_p(z) = \phi(z) G_p(z), \tag{15}$$

where the map $z \mapsto G_p(z)$ is non-decreasing by Lemma 6. Define $\bar{z} := (1/\underline{\sigma} - 1/\overline{\sigma})x$, so that $\mu_j \leq \bar{z}$ for every $1 \leq j \leq p$. Moreover, define $\bar{Z} := \max_{1\leq j\leq p}(W_j - \mu_j)$. Then

$$\begin{aligned}\int_z^\infty \phi(u)du G_p(z) \leq \int_z^\infty \phi(u) G_p(u)du &= \mathrm{P}(Z > z) \\ &\leq \mathrm{P}(\bar{Z} > z - \bar{z}) \leq \exp\left\{-\frac{(z - \bar{z} - \mathrm{E}[\bar{Z}])_+^2}{2}\right\},\end{aligned}$$

where the last inequality is due to the Gaussian concentration inequality (Lemma 7). Note that $W_j - \mu_j = X_j/\sigma_j$, so that

$$\mathrm{E}[\bar{Z}] = \mathrm{E}[\max_{1\leq j\leq p} (X_j/\sigma_j)] =: a_p.$$

Therefore, for every $z \in \mathbb{R}$,

$$G_p(z) \leq \frac{1}{1 - \Phi(z)} \exp\left\{-\frac{(z - \bar{z} - a_p)_+^2}{2}\right\}. \tag{16}$$

Mill's inequality states that for $z > 0$,

$$z \leq \frac{\phi(z)}{1 - \Phi(z)} \leq z\frac{1 + z^2}{z^2},$$

and in particular $(1 + z^2)/z^2 \leq 2$ when $z > 1$. Moreover, $\phi(z)/\{1 - \Phi(z)\} \leq 1.53 \leq 2$ for $z \in (-\infty, 1)$. Therefore,

$$\phi(z)/\{1 - \Phi(z)\} \leq 2(z \vee 1), \ \forall z \in \mathbb{R}.$$

Hence we conclude from this, (16), and (15) that

$$f_p(z) \leq 2(z \vee 1) \exp\left\{-\frac{(z - \bar{z} - a_p)_+^2}{2}\right\}, \ \forall z \in \mathbb{R}.$$

Step 3. By Step 2, for every $y \in \mathbb{R}$ and $t > 0$, we have

$$\mathrm{P}\left(|Z - y| \leq t\right) = \int_{y-t}^{y+t} f_p(z)dz \leq 2t \max_{z\in[y-t,y+t]} f_p(z) \leq 4t(\bar{z} + a_p + 1),$$

where the last inequality follows from the fact that the map $z \mapsto ze^{-(z-a)^2/2}$ (with $a > 0$) is non-increasing on $[a+1, \infty)$. Combining this inequality with Step 1, for every $x \geq 0$ and $\epsilon > 0$, we have

$$\mathrm{P}(|\max_{1\leq j\leq p} X_j - x| \leq \epsilon) \leq 4\epsilon\{(1/\underline{\sigma} - 1/\overline{\sigma})|x| + a_p + 1\}/\underline{\sigma}. \tag{17}$$

This inequality also holds for $x < 0$ by the similar argument, and hence it holds for every $x \in \mathbb{R}$.

If $\underline{\sigma} = \overline{\sigma} = \sigma$, then we have

$$\mathrm{P}(|\max_{1\leq j\leq p} X_j - x| \leq \epsilon) \leq 4\epsilon(a_p + 1)/\sigma, \ \forall x \in \mathbb{R}, \ \forall \epsilon > 0,$$

which leads to the first assertion of the theorem.

On the other hand, consider the case where $\underline{\sigma} < \overline{\sigma}$. Suppose first that $0 < \epsilon \leq \underline{\sigma}$. By the Gaussian concentration inequality (Lemma 7), for $|x| \geq \epsilon + \overline{\sigma}(a_p + \sqrt{2\log(\underline{\sigma}/\epsilon)})$, we have

$$\begin{aligned}\mathrm{P}(|\max_{1\leq j\leq p} X_j - x| \leq \epsilon) &\leq \mathrm{P}(\max_{1\leq j\leq p} X_j \geq |x| - \epsilon) \\ &\leq \mathrm{P}\left(\max_{1\leq j\leq p} X_j \geq \mathrm{E}[\max_{1\leq j\leq p} X_j] + \overline{\sigma}\sqrt{2\log(\underline{\sigma}/\epsilon)}\right) \leq \epsilon/\underline{\sigma}. \end{aligned} \tag{18}$$

For $|x| \leq \epsilon + \overline{\sigma}(a_p + \sqrt{2\log(\underline{\sigma}/\epsilon)})$, by (17) and using $\epsilon \leq \underline{\sigma}$, we have

$$\begin{aligned}&\mathrm{P}(|\max_{1\leq j\leq p} X_j - x| \leq \epsilon) \\ &\quad \leq 4\epsilon\{(\overline{\sigma}/\underline{\sigma})a_p + (\overline{\sigma}/\underline{\sigma} - 1)\sqrt{2\log(\underline{\sigma}/\epsilon)} + 2 - \underline{\sigma}/\overline{\sigma}\}/\underline{\sigma}. \end{aligned} \tag{19}$$

Combining (18) and (19), we obtain the inequality in (ii) for $0 < \epsilon \leq \underline{\sigma}$ (with a suitable choice of C). If $\epsilon > \underline{\sigma}$, the inequality in (ii) trivially follows by taking $C \geq 1/\underline{\sigma}$. This completes the proof. □

Proof of Lemma 5. Let $M := \max_{1\leq j\leq p} W_j$. The absolute continuity of the distribution of M is deduced from the fact that $\mathrm{P}(M \in A) \leq \sum_{j=1}^p \mathrm{P}(W_j \in A)$ for every Borel measurable subset A of $\mathbb{R}$. Hence, to show that a version of the density of M is given by (13), it is enough to show that $\lim_{\epsilon\downarrow 0} \epsilon^{-1}\mathrm{P}(x < M \leq x + \epsilon)$ equals the right side on (13) for a.e. $x \in \mathbb{R}$.

For every $x \in \mathbb{R}$ and $\epsilon > 0$, observe that

$$\begin{aligned}&\{x < M \leq x + \epsilon\} \\ &= \{\exists i_0, W_{i_0} > x \text{ and } \forall i, W_i \leq x + \epsilon\} \\ &= \{\exists i_1, x < W_{i_1} \leq x + \epsilon \text{ and } \forall i \neq i_1, W_i \leq x\} \\ &\quad \cup \{\exists i_1, \exists i_2, x < W_{i_1} \leq x + \epsilon, x < W_{i_2} \leq x + \epsilon \text{ and } \forall i \notin \{i_1, i_2\}, W_i \leq x\} \\ &\qquad \vdots \\ &\quad \cup \{\forall i, x < W_i \leq x + \epsilon\} \\ &=: A_1^{x,\epsilon} \cup A_2^{x,\epsilon} \cup \cdots \cup A_p^{x,\epsilon}. \end{aligned}$$

Note that the events $A_1^{x,\epsilon}, A_2^{x,\epsilon}, \ldots, A_p^{x,\epsilon}$ are disjoint. For $A_1^{x,\epsilon}$, since

$$A_1^{x,\epsilon} = \cup_{i=1}^p \{x < W_i \leq x + \epsilon \text{ and } W_j \leq x, \forall j \neq i\},$$

where the events on the right side are disjoint, we have

$$\begin{aligned} \mathrm{P}(A_1^{x,\epsilon}) &= \sum_{i=1}^p \mathrm{P}(x < W_i \leq x + \epsilon \text{ and } W_j \leq x, \forall j \neq i) \\ &= \sum_{i=1}^p \int_x^{x+\epsilon} \mathrm{P}(W_j \leq x, \forall j \neq i \mid W_i = u)\phi(u - \mu_i)du, \end{aligned}$$

where $\mu_i := \mathrm{E}[W_i]$. We show that for every $1 \leq i \leq p$ and a.e. $x \in \mathbb{R}$, the map $u \mapsto \mathrm{P}(W_j \leq x, \forall j \neq i \mid W_i = u)$ is right continuous at x. Let $X_j = W_j - \mu_j$ so that X_j are standard Gaussian random variables. Then

$$\mathrm{P}(W_j \leq x, \forall j \neq i \mid W_i = u) = \mathrm{P}(X_j \leq x - \mu_j, \forall j \neq i \mid X_i = u - \mu_i).$$

Pick $i = 1$. Let $V_j = X_j - \mathrm{E}[X_j X_1]X_1$ be the residual from the orthogonal projection of X_j on X_1. Note that the vector $(V_j)_{2 \leq j \leq p}$ and X_1 are jointly Gaussian and uncorrelated, and hence independent, by which we have

$$\begin{aligned} &\mathrm{P}(X_j \leq x - \mu_j, 2 \leq \forall j \leq p \mid X_1 = u - \mu_1) \\ &= \mathrm{P}(V_j \leq x - \mu_j - \mathrm{E}[X_j X_1](u - \mu_1), 2 \leq \forall j \leq p \mid X_1 = u - \mu_1) \\ &= \mathrm{P}(V_j \leq x - \mu_j - \mathrm{E}[X_j X_1](u - \mu_1), 2 \leq \forall j \leq p). \end{aligned}$$

Define $J := \{j \in \{2, \ldots, p\} : \mathrm{E}[X_j X_1] \leq 0\}$ and $J^c := \{2, \ldots, p\} \backslash J$. Then

$$\begin{aligned} &\mathrm{P}(V_j \leq x - \mu_j - \mathrm{E}[X_j X_1](u - \mu_1), 2 \leq \forall j \leq p) \\ &\to \mathrm{P}(V_j \leq x_j, \forall j \in J, V_{j'} < x_{j'}, \forall j' \in J^c), \text{ as } u \downarrow x, \end{aligned}$$

where $x_j = x - \mu_j - \mathrm{E}[X_j X_1](x - \mu_1)$. Here each V_j either degenerates to 0 (which occurs only when X_j and X_1 are perfectly negatively correlated, that is, $\mathrm{E}[X_j X_1] = -1$) or has a non-degenerate Gaussian distribution, and hence for every $x \in \mathbb{R}$ expect for at most $(p-1)$ points $(\mu_1 + \mu_j)/2, 2 \leq j \leq p$,

$$\begin{aligned} \mathrm{P}(V_j \leq x_j, \forall j \in J, V_{j'} < x_{j'}, \forall j' \in J^c) &= \mathrm{P}(V_j \leq x_j, 2 \leq \forall j \leq p) \\ &= \mathrm{P}(W_j \leq x, 2 \leq \forall j \leq p \mid W_1 = x). \end{aligned}$$

Hence for $i = 1$ and a.e. $x \in \mathbb{R}$, the map $u \mapsto \mathrm{P}(W_i \leq x, \forall j \neq i \mid W_i = u)$ is right continuous at x. The same conclusion clearly holds for $2 \leq i \leq p$. Therefore, we conclude that, for a.e. $x \in \mathbb{R}$, as $\epsilon \downarrow 0$,

$$\begin{aligned} \frac{1}{\epsilon}\mathrm{P}(A_1^{x,\epsilon}) &\to \sum_{i=1}^p \mathrm{P}(W_j \leq x, \forall j \neq i \mid W_i = x)\phi(x - \mu_i) \\ &= \phi(x) \sum_{i=1}^p e^{\mu_i x - \mu_i^2/2} \mathrm{P}(W_j \leq x, \forall j \neq i \mid W_i = x). \end{aligned}$$

In the rest of the proof, we show that, for every $2 \leq i \leq p$ and $x \in \mathbb{R}$, $\mathrm{P}(A_i^{x,\epsilon}) = o(\epsilon)$ as $\epsilon \downarrow 0$, which leads to the desired conclusion. Fix any $2 \leq i \leq p$. The probability $\mathrm{P}(A_i^{x,\epsilon})$ is bounded by a sum of terms of

the form $\mathrm{P}(x < W_j \le x+\epsilon, x < W_k \le x+\epsilon)$ with $j \neq k$. Recall that $\mathrm{Corr}(W_j, W_k) < 1$. Assume that $\mathrm{Corr}(W_j, W_k) = -1$. Then for every $x \in \mathbb{R}$, $\mathrm{P}(x < W_j \le x+\epsilon, x < W_k \le x+\epsilon)$ is zero for sufficiently small ϵ. Otherwise, $(W_j, W_k)^T$ obeys a two-dimensional, non-degenerate Gaussian distribution and hence $\mathrm{P}(x < W_j \le x+\epsilon, x < W_k \le x+\epsilon) = O(\epsilon^2) = o(\epsilon)$ as $\epsilon \downarrow 0$ for every $x \in \mathbb{R}$. This completes the proof. □

Proof of Lemma 6. Since $\mathrm{E}[W_0] \ge 0$, the map $x \mapsto \exp(\mathrm{E}[W_0]x - (\mathrm{E}[W_0])^2)$ is non-decreasing. Thus it suffices to show that the map

$$x \mapsto \mathrm{P}(W_1 \le x, \ldots, W_p \le x \mid W_0 = x) \tag{20}$$

is non-decreasing. As in the proof of Lemma 5, let $X_j = W_j - \mathrm{E}[W_j]$ and let $V_j = X_j - \mathrm{E}[X_j X_0]X_0$ be the residual from the orthogonal projection of X_j on X_0. Note that the vector $(V_j)_{1\le j\le p}$ and X_0 are independent. Hence the probability in (20) equals

$$\begin{aligned}&\mathrm{P}(V_j \le x - \mu_j - \mathrm{E}[X_j X_0](x - \mathrm{E}[W_0]), 1 \le \forall j \le p \mid X_0 = x - \mathrm{E}[W_0])\\ &= \mathrm{P}(V_j \le x - \mu_j - \mathrm{E}[X_j X_0](x - \mathrm{E}[W_0]), 1 \le \forall j \le p),\end{aligned}$$

where the latter is non-decreasing in x on $\mathbb{R}$ since $\mathrm{E}[X_j X_0] \le 1$. □

Appendix A. Proof of Lemma 1

Lemma 1 follows from the following maximal inequality and Hölder's inequality. Here we write $a \lesssim b$ if a is smaller than or equal to b up to a universal positive constant.

Lemma 8. *Let $Z_1, \ldots, Z_n$ be independent random vectors in $\mathbb{R}^p$ with $p \ge 2$. Define $M := \max_{1\le i\le n} \max_{1\le j\le p} |Z_{ij}|$ and $\sigma^2 := \max_{1\le j\le p} \sum_{i=1}^n \mathrm{E}[Z_{ij}^2]$. Then*

$$\mathrm{E}[\max_{1\le j\le p} |\textstyle\sum_{i=1}^n (Z_{ij} - \mathrm{E}[Z_{ij}])|] \lesssim (\sigma\sqrt{\log p} + \sqrt{\mathrm{E}[M^2]}\log p).$$

We shall use the following lemma.

Lemma 9. *Let $V_1, \ldots, V_n$ be independent random vectors in $\mathbb{R}^p$ with $p \ge 2$ such that $V_{ij} \ge 0$ for all $1 \le i \le n$ and $1 \le j \le p$. Then*

$$\mathrm{E}[\max_{1\le j\le p} \textstyle\sum_{i=1}^n V_{ij}] \lesssim \max_{1\le j\le p} \mathrm{E}[\sum_{i=1}^n V_{ij}] + \mathrm{E}[\max_{1\le i\le n} \max_{1\le j\le p} V_{ij}] \log p.$$

Proof of Lemma 9. We make use of the symmetrization technique. Let $\varepsilon_1, \ldots, \varepsilon_n$ be independent Rademacher random variables (that is, $\mathrm{P}(\varepsilon_i = 1) = \mathrm{P}(\varepsilon_i = -1) = 1/2$) independent of $V_1^n := \{V_1, \ldots, V_n\}$. Then by the triangle inequality and Lemma 2.3.1 in [28],

$$\begin{aligned}I := \mathrm{E}[\max_{1\le j\le p} \textstyle\sum_{i=1}^n V_{ij}] &\le \max_{1\le j\le p} \mathrm{E}[\textstyle\sum_{i=1}^n V_{ij}] + \mathrm{E}[\max_{1\le j\le p} |\textstyle\sum_{i=1}^n (V_{ij} - \mathrm{E}[V_{ij}])|]\\ &\le \max_{1\le j\le p} \mathrm{E}[\textstyle\sum_{i=1}^n V_{ij}] + 2\mathrm{E}[\max_{1\le j\le p} |\textstyle\sum_{i=1}^n \varepsilon_i V_{ij}|].\end{aligned}$$

By Lemmas 2.2.2 and 2.2.7 in [28], we have

$$\begin{aligned}\mathrm{E}[\max_{1\le j\le p}|\textstyle\sum_{i=1}^n \varepsilon_i V_{ij}| \mid V_1^n] &\lesssim \max_{1\le j\le p}(\textstyle\sum_{i=1}^n V_{ij}^2)^{1/2}\sqrt{\log p}\\ &\le \sqrt{B\log p}\max_{1\le j\le p}(\textstyle\sum_{i=1}^n V_{ij})^{1/2},\end{aligned}$$

where $B := \max_{1\le i\le n}\max_{1\le j\le p} V_{ij}$. Hence by Fubini's theorem and the Cauchy-Schwarz inequality,

$$\begin{aligned}\mathrm{E}[\max_{1\le j\le p}|\textstyle\sum_{i=1}^n \varepsilon_i V_{ij}|] &\lesssim \sqrt{\mathrm{E}[B]\log p}(\mathrm{E}[\max_{1\le j\le p}\textstyle\sum_{i=1}^n V_{ij}])^{1/2}\\ &= \sqrt{\mathrm{E}[B]\log p}\sqrt{I}.\end{aligned}$$

Therefore, we have

$$I \lesssim \max_{1\le j\le p}\mathrm{E}[\textstyle\sum_{i=1}^n V_{ij}] + \sqrt{\mathrm{E}[B]\log p}\sqrt{I} =: a + b\sqrt{I}.$$

Solving this inequality, we conclude that $I \lesssim a + b^2$. □

Proof of Lemma 8. Let $\varepsilon_1,\dots,\varepsilon_n$ be independent Rademacher random variables independent of $Z_1,\dots,Z_n$. Then arguing as in the previous proof, we have

$$\begin{aligned}\mathrm{E}[\max_{1\le j\le p}|\textstyle\sum_{i=1}^n (Z_{ij}-\mathrm{E}[Z_{ij}])|] &\le 2\mathrm{E}[\max_{1\le j\le p}|\textstyle\sum_{i=1}^n \varepsilon_i Z_{ij}|]\\ &\lesssim \mathrm{E}[\max_{1\le j\le p}(\textstyle\sum_{i=1}^n Z_{ij}^2)^{1/2}]\sqrt{\log p}\\ &\le (\mathrm{E}[\max_{1\le j\le p}\textstyle\sum_{i=1}^n Z_{ij}^2])^{1/2}\sqrt{\log p}. \quad \text{(Jensen)}\end{aligned}$$

By Lemma 9 applied to $V_{ij} = Z_{ij}^2$, we have

$$\mathrm{E}[\max_{1\le j\le p}\textstyle\sum_{i=1}^n Z_{ij}^2] \lesssim \sigma^2 + \mathrm{E}[M^2]\log p.$$

This implies the desired conclusion. □

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